Qubits and axioms of quantum computing

4

We now start to discuss the basic of quantum computing. In this chapter we introduce the definition of qubits, the axioms of quantum computing and we consider a first application of quantum computing: quantum key distribution. Then we also discuss systems made of several qubits.

I What's a qubit ?

In (classical) information theory, the basic unit of information is the *bit* of information which is either 0 or 1. In quantum information, the basic unit of information will be the *quantum bit* or *qubit*.

I.1 The quantum bit

We begin with a formal definition of what a qubit is and then we give examples coming from physics to provide an intuition about where qubits can be found.

Definition 23. A qubit is a normalized vector of $\mathbb{C}^2$. More precisely, using the Dirac notation, if we denote by $|0\rangle = \begin{pmatrix}1\\0\end{pmatrix}$ and $|1\rangle = \begin{pmatrix}0\\1\end{pmatrix}$ an orthonormal basis of $\mathbb{C}^2$, then a qubit $|\psi\rangle$ is a vector,

$$|\psi\rangle = a\,|0\rangle + b\,|1\rangle \text{ with } |a|^2 + |b|^2 = 1. \tag{4.1}$$

In the definition normalized means normalized for the hermitian inner product, i.e. the inner product over the complex numbers. Let v and w be two complex vectors of $\mathbb{C}^n$ with complex coordinates v_i and w_i, $i = 1,\dots,n$, then

$$\langle v,w\rangle = \overline{v}_1 w_1 + \cdots + \overline{v_n} w_n \tag{4.2}$$

The hermitian inner product satisfies the following properties,

- $\langle v,w\rangle = \overline{\langle w,v\rangle}$
- $\langle v,aw+bu\rangle = a\langle v,w\rangle + b\langle v,u\rangle$ and $\langle av+bu,w\rangle = \overline{a}\langle v,w\rangle + \overline{b}\langle u,w\rangle$.
- $\langle v,v\rangle \geq 0$ and $\langle v,v\rangle = 0 \Rightarrow v = 0_{\mathbb{C}^n}$.

Remark 9. Notations. We will use in these notes the bra/ket Dirac notation to represent vectors, linear applications and inner product. Vectors will be denoted with the *ket* notation, i.e. if $w = \begin{pmatrix}w_1\\w_2\\\vdots\\w_n\end{pmatrix} \in \mathbb{C}^n$ we will denote it by $|w\rangle$. The conjugate transpose of a vector v, i.e. $v^\dagger = {}^t\overline{v} = (\overline{v_1},\overline{v_2},\dots,\overline{v_n})$ will be denoted with the *bra* symbol, i.e. $w^\dagger = \langle w|$. This notation will be very convenient to perform calculations. For example the inner product of v and w is just the bra-ket product:

$$\langle v,w\rangle = \langle v|w\rangle = (\overline{v_1},\overline{v_2},\dots,\overline{v_n}).\begin{pmatrix}w_1\\w_2\\\vdots\\w_n\end{pmatrix} = \sum_{i=1}^{n} \overline{v_i} w_i \tag{4.3}$$

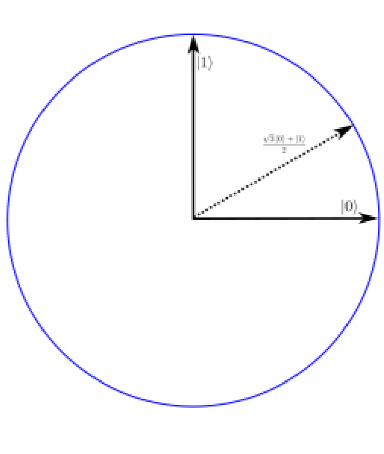


Figure 4.1: A qubit: $\left|\psi\right\rangle = \frac{\sqrt{3}}{2}\left|0\right\rangle + \frac{1}{2}\left|1\right\rangle$.

In classical information, a bit of information b has a definite value. In quantum information that is not the case because quantum physics is probabilistic. To obtain the value we need to measure the qubit.

Definition 24. Let $\left|\psi\right\rangle = a\left|0\right\rangle + b\left|1\right\rangle$ be a qubit expressed in the standard basis $(\left|0\right\rangle, \left|1\right\rangle)$. Measuring the qubit in the $(\left|0\right\rangle, \left|1\right\rangle)$ basis consists in projecting $\left|\psi\right\rangle$ in that basis.

$$\begin{tikzcd}
& \left|\psi\right\rangle \arrow[dl, squiggly, "|a|^2"'] \arrow[dr, squiggly, "|b|^2"] & \\
\left|0\right\rangle & & \left|1\right\rangle
\end{tikzcd} \tag{4.4}$$

The outcomes of that projection is probabilistic and the squared modules of the amplitudes $|a|^2$ and $|b|^2$ correspond to the probabilities of observing $\left|0\right\rangle$ or $\left|1\right\rangle$. After the measurement the state $\left|\psi\right\rangle$ is either $\left|0\right\rangle$ or $\left|1\right\rangle$.

Remark 10. In the definition we choose to measure the qubit in the standard basis but that same process can be performed in any orthonormal basis.

Example 1. An other basis of interest that we will use is the Hadamard basis

$$\left|+\right\rangle = \frac{1}{\sqrt{2}}\left|0\right\rangle + \frac{1}{\sqrt{2}}\left|1\right\rangle \text{ and } \left|-\right\rangle = \frac{1}{\sqrt{2}}\left|0\right\rangle - \frac{1}{\sqrt{2}}\left|1\right\rangle \tag{4.5}$$

1 Express the vectors of the standard basis $\left|0\right\rangle$, $\left|1\right\rangle$ in the Hadamard basis (i.e. in terms of $\left|+\right\rangle$ and $\left|-\right\rangle$).

I.2 The Stern-Gerlach experiment

We now introduce the Stern-Gerlach experiment to provide physical motivation and intuition regarding the existence of qubits as well as justifying the formal definition of a qubit. This experience was conducted in 1915. Originally Stern and Gerlach wanted to observe experimentally the intrinsic orientation of the angular momentum of silver atoms. They observed, Figure 4.2, that the atoms sent trough a magnetic field are only deflected along two directions North/South with respect to the orientation of the magnetic field. If particles behaved like *little magnet* one would expect to observe a continuum of possible orientations. This experience is considered as the first observation of a new physical quantity: the spin of a particle. The spin is a quantized quantity attached to particles and it is a two level quantum system for particles of type $\frac{1}{2}$-spin like electrons.

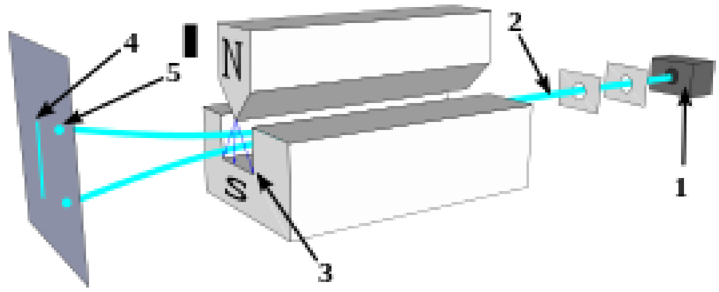


Figure 4.2: A beam of electrons is sent through a magnetic field. Only two types of deflection North/South is observed and not a continuous set of deflection.

In the language of quantum physics, the result of the experiment can be explained as follow. If we accept that the spin of an electron is quantized and has two basis states we can write the state of the spin of the particle as

$$\left|\psi_{\text{spin}}\right\rangle = a\left|\uparrow\right\rangle + b\left|\downarrow\right\rangle. \tag{4.6}$$

This is a linear combination of two basis wave functions $|\uparrow\rangle$ and $|\downarrow\rangle$. Therefore according to Definiton 24 the positive numbers $|a|^2$ and $|b|^2$ represent the probabilities of observing the states $|\uparrow\rangle$ or $|\downarrow\rangle$. For instance if $\left|\psi_{\text{spin}}\right\rangle = \frac{\sqrt{2}}{2}|\uparrow\rangle + \frac{\sqrt{2}}{2}|\downarrow\rangle$ then there will be a probability $\frac{1}{2}$ to observe a deflection to the North or to the South.

More interestingly in this experiment is what happens when we can combine several Stern-Gerlach apparatus in different directions (Figure 4.3).

In the x-direction let us denote the basis states by $|\rightarrow\rangle = \frac{1}{\sqrt{2}}(|\uparrow\rangle + |\downarrow\rangle)$ and $|\leftarrow\rangle = \frac{1}{\sqrt{2}}(|\uparrow\rangle - |\downarrow\rangle)$.

Assume that the source produces electron whose quantum states are $\left|\psi_{\text{spin}}\right\rangle = \frac{1}{\sqrt{2}}(|\uparrow\rangle + |\downarrow\rangle)$. Then the experience shows that in the 3 settings of Figure 4.3, the probability of observing (after the last SG apparatus):

- (setting 1) $|\uparrow\rangle$ is 0.5 and 0.5 of observing nothing,
- (setting 2) $|\rightarrow\rangle$ is 0.25, $|\leftarrow\rangle$ is 0.25 and 0.5 of observing nothing,

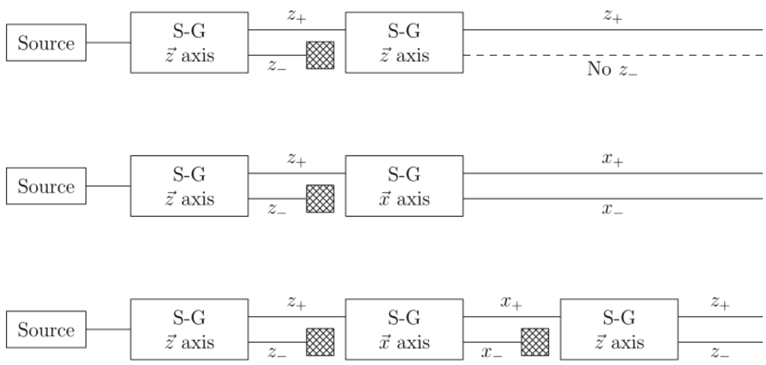


Figure 4.3: A sequence of Stern-Gerlach apparatus, the electron sent is blocked in one of the possible outcomes.

- (setting 3) $|\uparrow\rangle$ is 0.125, $|\downarrow\rangle$ is 0.125 and 0.75 of observing nothing.

This can be explained by an other feature of quantum mechanics: the state of a quantum system is modified by the measurement.

2 Use the formalism given by Definitions 23 and 24 to recover the probabilities observed for the multiple Stern-Gerlach experiment.

Remark 11. Other paradoxical experiments have nice and «simple» interpretation in the language of quantum physics. Those experiments are helpful to get an intuition about the three different features of quantum physics that will appear in quantum computation: *the superposition principle*, *the probabilistic issue of a measurement* and *the modification of the state after measurement*. Alternative experiences to explore those properties are: the double slits experiment, light beam and polarizers [10, 11, 12].

II The Axioms of quantum computation

We now formalize the rules or axioms based on the laws of quantum physics that we will use to perform quantum computation.

Axiom 25 (Superposition principle)

A quantum states $|\psi\rangle$ is a vector in a complex vector space equipped with an inner product. This complex vector space is called the Hilbert space $\mathscr{H}$ of the system. In particular if $|e_1\rangle, \ldots, |e_n\rangle$ is a basis of $\mathscr{H}$, the state $|\psi\rangle$ can be expressed as a linear combination of the states $|e_i\rangle$:

$$|\psi\rangle = a_1 |e_1\rangle + \ldots a_n |e_n\rangle . \tag{4.7}$$

Here the basis is orthonormal, i.e. $\langle e_i, e_i\rangle = 1$ and $\langle e_i, e_j\rangle = 0$ if $i \neq j$.

Axiom 26 (Measurement)

In equation (4.7) the complex amplitudes a_i satisfy

$$|a_1|^2 + \cdots + |a_n|^2 = 1 \tag{4.8}$$

and $|a_i|^2$ represents the probability of observing $|e_i\rangle$ when measuring the state $|\psi\rangle$ in the $|e_1\rangle, \ldots, |e_n\rangle$ basis.

Axiom 27 (Post-measurement)

After measurement the state $|\psi\rangle$ is in one of the basis states $|e_1\rangle, \ldots, |e_n\rangle$ of the measurement basis.

$$|\psi\rangle \rightsquigarrow |e_i\rangle \text{ with probability } |a_i|^2 \tag{4.9}$$

Remark 12. It is important to notice that the measurement are associated to a choice of basis of the Hilbert space. For instance in the Stern-Gerlach experiment measuring the spin in the z-direction corresponds to measuring the state $|\psi\rangle$ is the $|\uparrow\rangle$ and $|\downarrow\rangle$ basis. While measuring the spin in the x-direction corresponds to measuring the state in the $|\rightarrow\rangle$, $|\leftarrow\rangle$ basis. One says in quantum physics that the state is projected onto the basis.

The intuition regarding the next axiom has not been emphasized in the presentation of the Stern-Gerlach experiment. It concerns how a quantum state can be transformed. Roughly speaking we will consider only transformations which transform a quantum state to a quantum state with respect to the inner product, i.e. if two quantum states are orthogonal, they should remain orthogonal if we apply the same transformation and the norm should be preserved.

Definition 28. A linear transformation $f : \mathcal{H} \to \mathcal{H}$ is said to be unitary if and only if

$$\langle f(v), f(w)\rangle = \langle v, w\rangle, \forall v, w \in \mathcal{H} \tag{4.10}$$

If U_f is a matrix encoding the linear transformation f in a basis then

$${}^t\overline{U}_f U_f = Id \tag{4.11}$$

Axiom 29 (Time evolution)

A quantum state $|\psi\rangle$ evolves in time according to a unitary transformation, i.e.

$$|\psi(t)\rangle = U(t)|\psi_0\rangle. \tag{4.12}$$

III Quantum key distribution: the BB84 protocol

In the following exercise we introduce a protocol based on the properties of quantum computation that allows Alice and Bob to share a secret key.

3 *The BB84 protocol* This exercise introduces a quantum protocol to exchange secret key (i.e. a string of 0 and 1) between two parties Alice and Bob. This protocol known as BB84 is due to Bennett and Brassard and was published in 1984 [2].

Alice and Bob want to build a secret key that they will use to do cryptographic communication. They can exchange qubits (send photon) and classical information (telephone call) through a quantum and a classical channel but both of them are not secure (i.e. an eavesdropper Eve can observe the exchanges). The protocol goes as follow.

- Alice generates a random string of bits of length N.
- She encodes each bit of her string with a qubit but choosing at random at each step one of the following two possibilities:
  - **a**. she encodes in the standard basis: $0 \to |0\rangle$ and $1 \to |1\rangle$,
  - **b**. or she encodes the bit in the Hadamard basis: $0 \to |+\rangle$ and $1 \to |-\rangle$.
- She sends each qubit to Bob through the quantum channel.
- For each qubit he received, Bob measures it at random in either the standard or the Hadamard basis and note the outcomes.
- Once Bob got all the qubits sent by Alice, they tell each other (on the phone) which basis has been used for each qubit.
- The secret key is the one formed by the qubits where both of them have used the same basis (to encode and to measure).

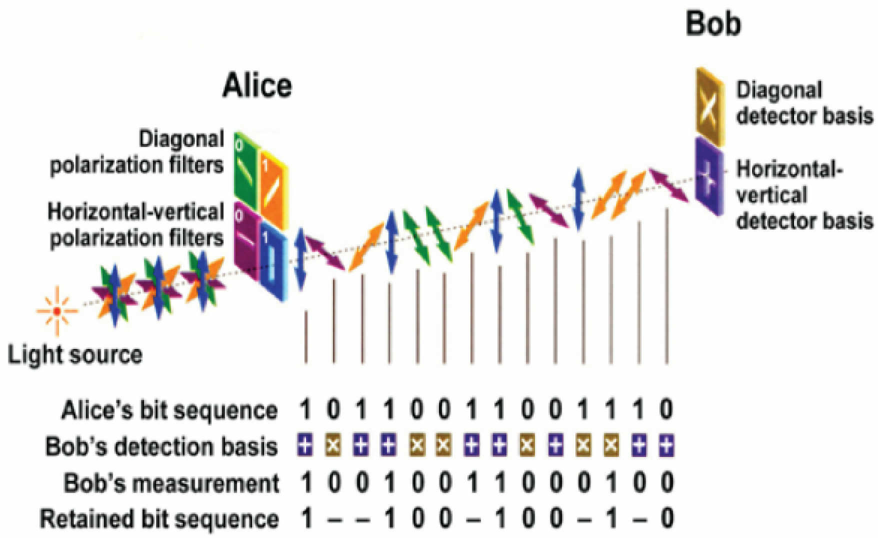
a. Suppose Alice started with the bit string 1010011010 and she randomly encodes it using the following basis (S for Standard, H for Hadamard), $SSHSHHSHSS$. Write the sequence of qubits which has been sent to Bob.

b. If Bob measures the qubits sent by Alice in the basis $HSHSSHHSHS$. What would be the shared key ?

c. If N is the number of qubits sent by Alice what is in average the number of qubits that may be used for the key (i.e. how many times will they get the same bit for sure)?

d. If Eve intercepts the qubits sent by Alice and make a measurement before sending it back to Bob what will happen ? What is the probability that Eve may observe the qubit without disturbing the protocol ?

Remark 13. Usually to complete the protocol, one add one more step where Alice and Bob compare for safety some bits of the share key. If they found the exact same bits they conclude that no eavesdropper was looking at the qubit exchange. The number of bits they agree to compare depends on the level of security they want to achieve.



IV Multipartite system and entanglement

Many interesting things start to happen when we consider several particles. First one explains mathematically how the Hilbert space of a composite system should be described. We review the axioms of chapter 4 in this new setting. Then one introduces the EPR paradox, the notion of entanglement and we end with a first quantum protocol: qubit teleportation.

IV.1 Tensor product

Let us consider two qubits (two particles) A and B which form a system $\left|\psi_{AB}\right\rangle \in \mathcal{H}$. How does $\mathcal{H}$ look like ? If we denote by $(|0_A\rangle, |1_A\rangle$ and $(|0_B\rangle, |1_B\rangle)$ the basis of the Hilbert spaces $\mathcal{H}_A = \mathcal{H}_B = \mathbb{C}^2$ then we would expect, according to the superposition principle, the state $\left|\psi_{AB}\right\rangle$ to be in a linear combination of $|0_A 0_B\rangle$ $|0_A 1_B\rangle$, $|1_A 0_B\rangle$ and $|1_A 1_B\rangle$, i.e.

$$\left|\psi_{AB}\right\rangle = a_{00}\,|0_A 0_B\rangle + a_{01}\,|0_A 1_B\rangle + a_{10}\,|1_A 0_B\rangle + a_{11}\,|1_A 1_B\rangle \tag{4.13}$$

with $|a_{00}|^2 + |a_{01}|^2 + |a_{10}|^2 + |a_{11}|^2 = 1$.

In other words the vector $\left|\psi_{AB}\right\rangle$ is part of a 4 dimensional vector space $\mathcal{H}_{AB}$ whose basis is obtained by «multiplying» the basis states of $\mathcal{H}_A$ and $\mathcal{H}_B$, i.e. $|00\rangle = |0\rangle \otimes |1\rangle, |01\rangle = |0\rangle \otimes |1\rangle, |10\rangle = |1\rangle \otimes |0\rangle$ and $|11\rangle = |1\rangle \otimes |1\rangle$ (for now on I drop the reference to the first particle A and second particle B).

This new vector space $\mathcal{H}_{AB}$ is in fact mathematically known as the tensor product of $\mathcal{H}_A$ and $\mathcal{H}_B$.

Definition 30. Let V and W two vector spaces equipped with basis $(v_1, \ldots, v_n)$ and $(w_1, \ldots, w_m)$. Then we call *the tensor product* of V and W the new vector space $V \otimes W$ of dimension nm spanned by the basis denoted by $v_i \otimes w_j$ with $1 \leq i \leq n$ and $1 \leq j \leq m$. In other words a vector of $u \in V \otimes W$ can be expressed as

$$u = \sum_{1\leq i\leq n, 1\leq j\leq m} \lambda_{ij} v_i \otimes w_j. \tag{4.14}$$

If V and W are equipped with inner products $\langle,\rangle_V$ and $\langle,\rangle_W$ then $V\otimes W$ is also equipped with an inner product

$$\langle v\otimes w, v'\otimes w'\rangle = \langle v,v'\rangle_V \langle w,w'\rangle_W. \tag{4.15}$$

(the definition extend by linearity). Thus if $(v_1,\dots,v_n)$ and $(w_1,\dots,w_m)$ are orthogonal basis of V and W, then $(v_1\otimes w_1,\dots,v_n\otimes w_m)$ is an orthogonal basis of $V\otimes W$.

Moreover we have the following properties,

- $(u+v)\otimes w = u\otimes w + v\otimes w$
- $u\otimes(w+z) = u\otimes w + u\otimes z$
- $\lambda u\otimes w = (\lambda u)\otimes w = u\otimes(\lambda w)$.

Remark 14. When dealing with qubits (or qudits) we will often use the short hand notation for the new basis $|i_1 i_2 \dots i_n\rangle = |i_1\rangle\otimes|i_2\rangle\otimes\dots\otimes|i_n\rangle$.

The Hilbert space of a two qubit system is therefore $\mathscr{H}_{AB} = \mathbb{C}^2\otimes\mathbb{C}^2$ and a general 2-qubit state can be written as:

$$\left|\psi_{AB}\right\rangle = a_{00}|00\rangle + a_{01}|01\rangle + a_{10}|10\rangle + a_{11}|11\rangle. \tag{4.16}$$

The basis $(|00\rangle,|01\rangle,|10\rangle,|11\rangle)$ is an orthonormal basis of $\mathscr{H}_{AB}$ (standard basis).

Similarly the Hilbert space of a three qubit system is

$$\mathscr{H}_{ABC} = \mathbb{C}^2\otimes\mathbb{C}^2\otimes\mathbb{C}^2. \tag{4.17}$$

and a general 3-qubit states can be written as

$$\begin{aligned}\left|\psi_{ABC}\right\rangle &= a_{000}|000\rangle + a_{001}|001\rangle + a_{010}|010\rangle + a_{100}|100\rangle \\ &\quad + a_{011}|011\rangle + a_{101}|101\rangle + a_{110}|110\rangle + a_{111}|111\rangle.\end{aligned} \tag{4.18}$$

with $\sum_{i,j,k\in\{0,1\}} |a_{ijk}|^2 = 1$.

Remark 15. One can see here one important feature of quantum computation through the structure of the space of a multipartite system. Namely if we are dealing with n qubits, the Hilbert space is given by $\mathscr{H} = \underbrace{\mathbb{C}^2\otimes\dots\otimes\mathbb{C}^2}_{n \text{ times}}$ which is a vector space of dimension 2^n. For instance if we were considering a 300 qubits quantum computer (i.e. a machine working with 300 2-level systems) then the computational space, i.e. the Hilbert space describing all possible configurations will have a dimension greater than the estimated number of observable particles in the universe...

IV.2 Measurement

In multipartite systems the measurement principles are similar to the one described in Chapter 4. Namely we choose an orthonormal basis of the Hilbert space and we express the states $|\psi\rangle$ in that basis. The square of the module of the amplitudes will give us the probabilities of observing one particular outcome and the state $|\psi\rangle$ will projected in one of the basis states after measurement.

Let $|\psi\rangle = \frac{1}{\sqrt{2}}|00\rangle + \frac{1}{\sqrt{2}}|11\rangle$ then if we measure that system in the standard basis, one has probability $\frac{1}{2}$ of observing $|00\rangle$ or $|11\rangle$ and probability 0 of observing $|01\rangle$ or $|10\rangle$.

Now in a composite system we can address the following question: what happens if we measure only one qubit ?

Let us consider the general two qubit state

$$\left|\psi_{AB}\right\rangle = a_{00}\left|00\right\rangle + a_{01}\left|01\right\rangle + a_{10}\left|10\right\rangle + a_{11}\left|11\right\rangle. \tag{4.19}$$

The probability of observing for example $|0\rangle$ for particle A is the sum of two independent events: observing $|00\rangle$ and $|01\rangle$. That is to say

$$p_{A=|0\rangle} = |a_{00}|^2 + |a_{01}|^2. \tag{4.20}$$

and similarly we have

$$p_{A=|1\rangle} = |a_{10}|^2 + |a_{11}|^2, p_{B=|0\rangle} = |a_{00}|^2 + |a_{10}|^2, p_{B=|1\rangle} = |a_{01}|^2 + |a_{11}|^2. \tag{4.21}$$

How would be the state $\left|\psi_{AB}\right\rangle$ affected if we measure the qubit A and observe $|0\rangle$? Well it has to be projected to the «part» of $\left|\psi_{AB}\right\rangle$ which will provide $|0\rangle$ for A, i.e.

$$\left|\psi_{AB}\right\rangle \rightsquigarrow \frac{a_{00}\left|00\right\rangle + a_{01}\left|01\right\rangle}{|a_{00}|^2 + |a_{01}|^2}. \tag{4.22}$$

(one divides by $|a_{00}|^2 + |a_{01}|^2$ to have a normalized state). The state is still in a superposition but the value of the qubit A is fixed because it has been measured. One defines similarly the post measurement states if we observed $A = |1\rangle$ or $B = |0\rangle, B = |1\rangle$.

IV.3 The EPR paradox & Entanglement

The two qubit case exhibits a nice quantum feature known as *quantum entanglement* which I describe now. Historically it begins with a paper of Einstein Podolsky and Rosen [7] in 1935 where the authors wanted to criticize the foundation of quantum mechanics. In particular Einstein was very unhappy with the probabilistic nature of quantum mechanics. According to Einstein quantum mechanics was incomplete, meaning that some "hidden variables", not yet introduced in the theory, should be added. If such variables could be found, they would allow to provide a deterministic version of quantum mechanics. To prove that the existence of hidden variables was necessary, the authors imagine the following «thought experiment»:

Assume you have a two particles system composed of spin $\frac{1}{2}$-particles. Then the superposition principle allows the following quantum state to exist:

$$\left|\psi_{EPR}\right\rangle = \frac{1}{\sqrt{2}}(|\uparrow\uparrow\rangle + |\downarrow\downarrow\rangle) = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle) \tag{4.23}$$

If such system, composed of two particles, exists one may imagine that the first particle, particle A, is sent to Alice and the second one, B, to Bob. One may also imagine that both Alice and Bob locations are far from each other (one on earth, the other on the moon). Then if Alice measures her qubit and get 0 then one knows according to the measurement principle that the state $\left|\psi_{EPR}\right\rangle$ has been projected to the only issue allowing Alice to measure 0, i.e. $\left|\psi_{EPR}\right\rangle \rightsquigarrow |00\rangle$ (similarly if she measured 1 one knows that $\left|\psi_{EPR}\right\rangle \rightsquigarrow |11\rangle$). Therefore after Alice measurement the state of the qubit hold by Bob is not in any superposition but completely determined (it is the same as Alice's one).

In their paper, Einstein, Podolsky and Rosen talked about *spooky action at the distance* because according to them everything was happening in the context of quantum mechanics as if information was sent instantaneously from Alice to Bob. Of course this last assertion was against the laws of special and general relativity. Thus for the authors, the solution to avoid this paradox (known as EPR paradox) is to assume that the theory is incomplete. With the addition of hidden variables, one should be able to predict deterministically the state of the qubit of Alice and Bob without assmuming any action at the distance (principle of *local realism*). The defender of quantum mechanics argue to Einstein that this was not a paradox at all. No information was exchanged. The measurement operated by Alice was still probabilistic (one does not know what the outcome will be) and what the EPR's paradox was pointing out was the possibility of correlations of quantum nature between spaced objects (non-locality).

The answer of this scientific controversy was provided some 50 years after Einstein et al.'s paper. First in 1966 John Bell provides inequalities on the possible outcomes of quantum measurement which were bounded by different values depending on the assumption of the existence of hidden variables or on the assumptions of quantum mechanics. Those inequalities, known as Bell's inequalities [3], opened the path to experimental tests for the EPR's paradox. This was first achieved, and reproduced many time since, by Alain Aspect in 1982 [1]. The results of Aspect's experience were agreeing with the assumptions of quantum mechanics and it was invalidating all potential hidden variables theories.

Let us go back to the maths... what is so special with this EPR quantum state $|\psi_{EPR}\rangle$?

Quantum states of the two-qubit Hilbert space $\mathcal{H} = \mathbb{C}^2 \otimes \mathbb{C}^2$ can be expressed as a matrix,

$$|\psi_{AB}\rangle = a_{00}|00\rangle + a_{01}|01\rangle + a_{10}|10\rangle + a_{11}|11\rangle = \begin{pmatrix} a_{00} & a_{01} \\ a_{10} & a_{11} \end{pmatrix}. \tag{4.24}$$

With this notation one remark that

$$|\psi_{EPR}\rangle = \begin{pmatrix} 1/\sqrt{2} & 0 \\ 0 & 1/\sqrt{2} \end{pmatrix}. \tag{4.25}$$

Let us now consider the state

$$|\psi_{Sep}\rangle = (a_A|0\rangle + b_A|1\rangle) \otimes (a_B|0\rangle + b_B|1\rangle). \tag{4.26}$$

This state has the property that measuring qubit A has no incidence on qubit B. If we obtain 0 for Alice, the state gets projected as $|\psi_{Sep}\rangle \rightsquigarrow |0\rangle \otimes (a_B|0\rangle + b_B|1\rangle)$ and thus B is still in a superposition. In a matrix form one obtains,

$$|\psi_{Sep}\rangle = \begin{pmatrix} a_A a_B & a_A b_B \\ b_A a_B & b_A b_B \end{pmatrix}. \tag{4.27}$$

If one compares (4.25) and (4.27) one may notice that the first matrix does not annihilate the determinant (the matrix has rank 2) while in the second case the determinant is zero (and in fact the matrix has rank 1). Another way of pointing out the difference of behavior between states $|\psi_{EPR}\rangle$ and $|\psi_{Sep}\rangle$ is to remark that $|\psi_{Sep}\rangle$ is «factorisable» while $|\psi_{EPR}\rangle$ is not.

Definition 31. A quantum state $|\psi\rangle$ of a multipartite Hilbert space $\mathcal{H} = \mathcal{H}_1 \otimes \cdots \otimes \mathcal{H}_n$ is said to be *separable* if and only if there exist states $|\psi_i\rangle \in \mathcal{H}_i$ such that

$$|\psi\rangle = |\psi_1\rangle \otimes \cdots \otimes |\psi_n\rangle. \tag{4.28}$$

If a state is not separable, it is called entangled.